Random Dice Roll with Charlieplexed LEDs

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Topic and Relevance:

This project demonstrates the implementation of Charlieplexing, a technique that allows multiple LEDs to be controlled using fewer pins on a microcontroller. This project creates a simple dice roller where a random number is generated and displayed on the Charlieplexed LEDs.

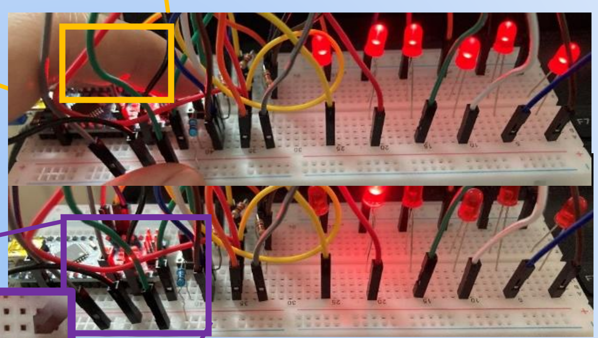
Program Code:

```
const int buttonPin = A0;
byte pins[] = {2,3,4};
const int NUMBER_OF_PINS = sizeof(pins)/ sizeof(pins[0]);
const int NUMBER_OF_LEDS = NUMBER_OF_PINS * (NUMBER_OF_PINS-1);
byte pairs[NUMBER_OF_LEDS/2][2] = { {0,1}, {1,2}, {0,2}};
void setup() { pinMode(buttonPin, INPUT_PULLUP); }
void loop() {
  if (digitalRead(buttonPin) == HIGH) {
    int i =random(0,NUMBER_OF_LEDS+1);
    lightLed(i); // calls the lightLed function to light the selected LED}}
void lightLed(int led){ // this function lights the given LED
  int indexA = pairs[led/2][0];
  int indexB = pairs[led/2][1];
```

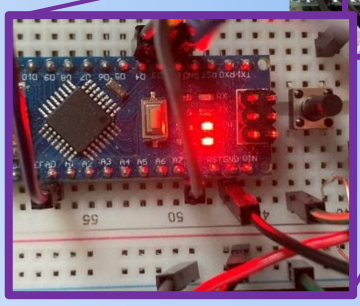
```
  int pinA = pins[indexA];
  int pinB = pins[indexB];
for(int i=0; i < NUMBER_OF_PINS; i++) {
        if(pins[i] != pinA && pins[i] != pinB){  // if this pin is not one of our pins
        pinMode(pins[i], INPUT); // set the mode to input
        digitalWrite(pins[i],LOW); }  // make sure pull-up is off
  pinMode(pinA, OUTPUT);
  pinMode(pinB, OUTPUT);
  if(led%2 == 0){
    digitalWrite(pinA,LOW);
    digitalWrite(pinB,HIGH);
  }else{
    digitalWrite(pinB,LOW);
    digitalWrite(pinA,HIGH);  } }
```

Results:

When the button is pressed



After the button is released



When the button is pressed, a random number between 1 and 6 is generated, and the corresponding LEDs are illuminated to represent the number.

Discussion and Outlook:

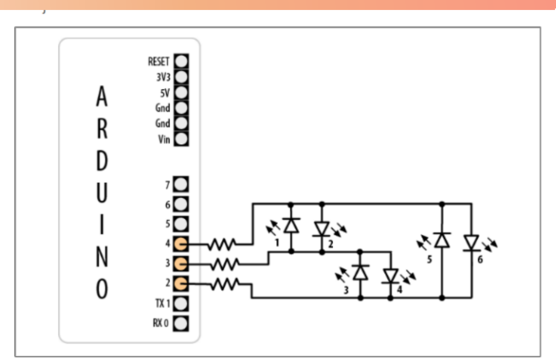


Figure 7-8. Six LEDs driven through three pins using Charlieplexing

Pins			LEDs					
2	3	4	1	2	3	4	5	6
L	L	L	0	0	0	0	0	0
L	H	i	1	0	0	0	0	0
H	L	i	0	1	0	0	0	0
i	L	H	0	0	1	0	0	0
i	H	L	0	0	0	1	0	0
L	i	H	0	0	0	0	1	0
H	i	L	0	0	0	0	0	1

Figure 7-8 from the Arduino Cookbook illustrates the Charlieplexing technique, where six LEDs are driven by only three pins.

The provided code effectively demonstrates the principles of Charlieplexing, a technique that allows multiple LEDs to be controlled using fewer pins on a microcontroller. The limitations of Charlieplexing, such as the number of LEDs, can be controlled and the potential for brightness variations.

As demonstrated in the provided image, it's possible to increase the number of controlled LEDs by adding more pins. By carefully planning the pin connections and LED pairings, you can create more complex patterns and displays.

References: Arduino Cookbook, 1st Edition by Michael Margolis. O'Reilly Media, 2011. Chapter 7.9: Controlling a Matrix of LEDs: Charlieplexing, pp. 239-242.